UF PHA 6125 — Clearance, Distribution, and Absorption

Based on UF PHA 6125 Module 2

Rowland/Tozer and Jeong/Jusko updates

Notes taken by NanFangHong

Module 2A/2B/2C 2026

Reading motive

这是 UF PHA 6125 的 Module 2 笔记。Module 2A 讲 renal and hepatic clearance，Module 2B 讲 protein binding and drug distribution，Module 2C 讲 oral drug absorption。我的目标不是把三份课件分别摘要，而是把它们改写成一张 *ADME* 系统图：drug 如何进入 systemic circulation，如何在 plasma/tissue 之间分布，如何被 liver 和 kidney 清除，以及这些机制如何进入 dose selection, DDI, organ impairment, food effect, bioavailability, and label thinking。

[course note]

Reader

假设读者刚学完 Module 1，对 CL, V_d, half-life, F, PK/PD 这些词有初步印象，但还没有真正把 physiology、protein binding、transporters、first pass、renal mechanisms 和 hepatic extraction 连起来。本笔记会尽量从读者视角解释每个公式背后的物理含义。

[from zero]

Update

课件来自 2020 年。本笔记保留 UF PHA 6125 的教学主线，同时参考 Rowland/Tozer Chapter 5 elimination，Jeong and Jusko 关于 f_d 和 classical hepatic clearance models 的更新，FDA renal/hepatic impairment 与 bioavailability guidance，以及 ICH M9/M12 对 absorption、bioequivalence、DDI、transporters and enzymes 的当前监管语言。

[2026 view]

Contents

1 Module Overview: ADME as One System

1.1 The three official module promises

Module 2A is titled *Clearance: renal and hepatic.* Its promise is to make CL concrete: clearance is the volume of plasma, blood, or unbound fluid cleared of drug per unit time, and the two major elimination routes are biotransformation and excretion. Biotransformation is usually enzymatic conversion of parent drug into metabolites, mostly in the liver but also in gut, kidney, lung, skin, and plasma. Excretion is removal of intact drug through urine, bile, saliva, body fluids, or milk.

Module 2B is titled *Protein binding and drug distribution.* Its promise is to explain why a measured plasma concentration is not the same thing as the amount of drug in the body or the concentration at a tissue target. Blood perfusion, plasma protein binding, tissue binding, regional pH, membrane permeability, and physicochemical properties all shape the rate and extent of distribution. V_d is the proportionality between amount in body and measured blood or plasma concentration; it is not an anatomical volume.

Module 2C is titled *Drug absorption.* Its promise is to explain how orally administered drug reaches systemic blood. Oral absorption is favored by the body acting as a sink: distribution and elimination keep systemic unbound concentration low, preserving a gradient across the gut wall. Gastric emptying, intestinal transit, dissolution, permeability, transporters, intestinal metabolism, hepatic first pass, degradation, and food can all alter rate and extent.

Taken together, these three modules form one statement:

> *The concentration we observe in plasma is the visible trace of input, distribution, and elimination all acting at once.*

This matters because almost every clinical pharmacology decision is an inverse problem. We give a dose and observe concentration–time data. From that trace we infer whether a molecule is dissolving, crossing membranes, binding to proteins, entering tissues, being extracted by liver, filtered or secreted by kidney, or being changed by food, disease, genotype, or interacting drugs.

1.2 Why the order feels strange at first

The course presents clearance first, then distribution, then absorption. Physiologically, a tablet obviously dissolves and absorbs before it is cleared. But pedagogically, clearance first is defensible. CL is the parameter that makes exposure interpretable:

$$AUC_{\mathrm{IV}} = \frac{D}{CL}, \qquad AUC_{\mathrm{oral}} = \frac{FD}{CL}.$$

If the denominator is not understood, the numerator cannot be interpreted. A low oral AUC may mean poor absorption, high first pass, high systemic clearance, or some combination of all three. Module 2 therefore starts by sharpening the exit term, then comes back to the distribution space and the oral input pathway.

Remark. I visually inspected the three original slide decks and transcript-derived flow diagrams before redrawing the figures below. The figures are not screenshots. They are compact LaTeX/TikZ reconstructions designed to survive both ordinary PDF rendering and the pdf2htmlEX HTML route used by this site.

1.3 One ADME diagram

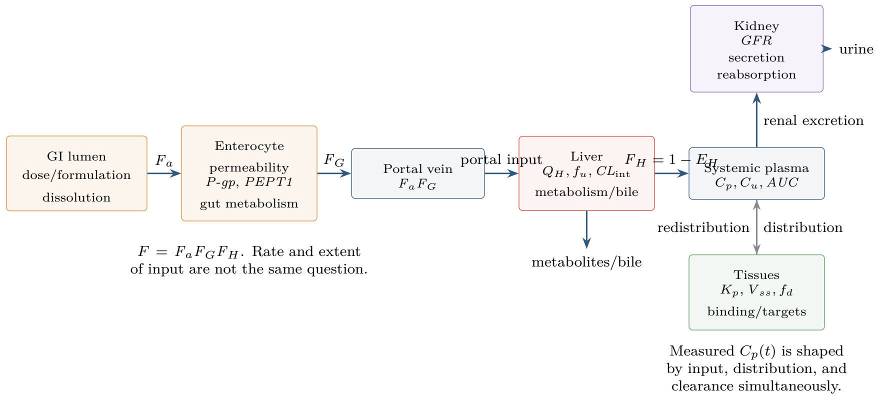

Figure 1: Module 2 in one picture. The apparent simplicity of $C_p(t)$ hides a coupled physiological system: gut input, hepatic first pass, distribution, renal clearance, hepatic clearance, protein binding, transport, and tissue partitioning.

1.4 The four concentration languages

The same molecule can be described by several concentrations. Confusing these is the fastest way to make *PK* formulas feel mystical.

$$\text{rate of elimination} = CL_pC_p = CL_BC_b = CL_uC_u.$$

The rate is physical. The clearance value depends on the concentration used as reference. This is why Rowland/Tozer repeatedly distinguishes plasma clearance, blood clearance, and unbound clearance [5]. When estimating organ extraction, the denominator should be blood flow and blood concentration, not plasma flow and plasma concentration:

$$CL_B = Q\,E, \qquad E = \frac{C_{\text{in}} - C_{\text{out}}}{C_{\text{in}}}.$$

The symbol *CL* is therefore incomplete unless we know whether it is plasma-based, blood-based, unbound, renal, hepatic, systemic, or apparent after oral dosing.

2 2A: Clearance

2.1 Clearance is a proportionality, not a bucket

The safest definition is:

$$\text{rate of elimination} = CL \times C.$$

If C is plasma concentration, CL is plasma clearance; if C is blood concentration, CL is blood clearance; if C is unbound concentration, CL is unbound clearance. The unit is volume per time, but no literal volume of plasma is scooped out and washed clean. Clearance is a conversion factor between concentration and elimination rate.

For linear first-order elimination, CL is constant:

$$-\frac{\mathrm{d}A}{\mathrm{d}t} = CL\,C.$$

For a one-compartment IV bolus with $C = A/V_d$,

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -\frac{CL}{V_d}C = -k_{\mathrm{el}}C, \qquad t_{1/2} = \frac{\log 2}{k_{\mathrm{el}}} = \frac{\log 2\ V_d}{CL}.$$

This is the key Module 1 connection: half-life is not a primary elimination capacity. It is the consequence of CL and V_d.

2.2 First-order, zero-order, and capacity-limited elimination

The Module 2A objectives ask for first-order and zero-order elimination. These are not just curve shapes.

- **First-order elimination**: a constant fraction of the amount is removed per unit time. Rate increases in proportion to concentration. CL is constant.
- **Zero-order elimination**: a constant amount is removed per unit time. Rate is approximately independent of concentration. Apparent $CL = \text{rate}/C$ decreases as concentration increases.
- **Michaelis–Menten elimination**: the mechanistic bridge between the two:
$$\text{rate} = \frac{V_{\max}C}{K_m + C}.$$
When $C \ll K_m$, rate $\approx (V_{\max}/K_m)C$, so elimination looks first-order. When $C \gg K_m$, rate $\approx V_{\max}$, so elimination looks zero-order.

The clinical point is that a dose increase can be harmless under linear elimination and dangerous under saturable elimination. The parameter AUC no longer scales linearly with dose once CL becomes concentration-dependent.

2.3 Mass balance across an organ

For a clearing organ at steady state, the input rate is QC_{in} and the output rate is QC_{out}. The amount removed per time is

$$Q(C_{\text{in}} - C_{\text{out}}).$$

Dividing by QC_{in} gives extraction ratio:

$$E = \frac{Q(C_{\text{in}} - C_{\text{out}})}{QC_{\text{in}}} = \frac{C_{\text{in}} - C_{\text{out}}}{C_{\text{in}}}.$$

Dividing the elimination rate by inlet blood concentration gives organ blood clearance:

$$CL_B = Q\,E.$$

So the maximum possible blood clearance of an organ is its blood flow. This statement is simple but powerful. It tells us why high-extraction drugs are flow-limited: if the organ already removes almost everything delivered to it, increasing enzyme activity cannot extract much more than everything.

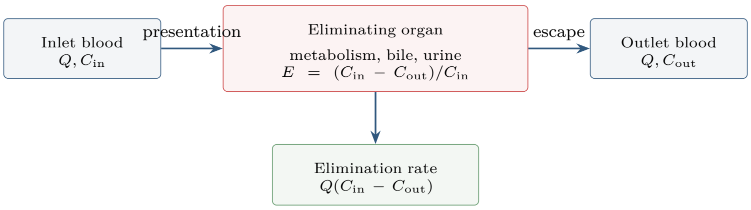

Figure 2: Organ clearance is a mass-balance idea. The extraction ratio is a fraction; blood clearance is blood flow times that fraction.

2.4 Total clearance is additive

If a drug is eliminated by several independent routes, rates add:

$$\text{rate}_{\text{total}} = \text{rate}_R + \text{rate}_H + \cdots.$$

For the same reference concentration, clearances add:

$$CL = CL_R + CL_H + \cdots.$$

This is why the fraction excreted unchanged in urine is so useful:

$$f_e = \frac{CL_R}{CL}.$$

If f_e is large, renal impairment can matter directly. If f_e is small, renal impairment can still matter indirectly through changes in transporters, metabolites, protein binding, or nonrenal clearance, but the first-order suspicion is different.

2.5 Renal clearance: filtration, secretion, reabsorption

Renal clearance decomposes into three processes:

$$\text{excretion} = \text{filtration} + \text{secretion} - \text{reabsorption}.$$

Only unbound drug is freely filtered at the glomerulus, so filtration clearance is

$$CL_{\text{filtration}} = f_u \, GFR.$$

In a typical adult, *GFR* is around 120–130 mL/min, while renal blood flow is much larger, around 1.1 L/min [5, 9]. Therefore:

$$CL_R \approx f_u\,GFR \quad \Rightarrow \quad \text{filtration-dominated};$$

$$CL_R > f_u\,GFR \quad \Rightarrow \quad \text{net active secretion};$$

$$CL_R < f_u\,GFR \quad \Rightarrow \quad \text{net reabsorption}.$$

This diagnostic rule is one of the highest-yield pieces of renal *PK*.

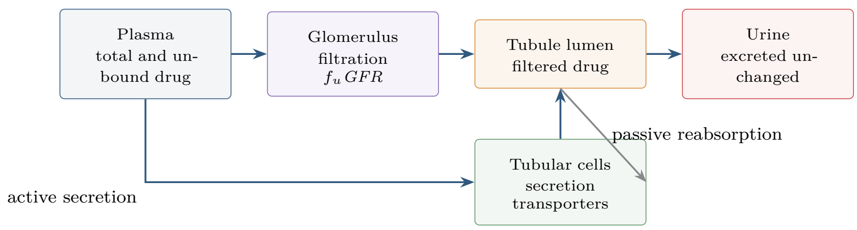

Figure 3: Renal clearance is not just filtration. Tubular secretion can make CL_R exceed $f_u\,GFR$; reabsorption can make it smaller.

2.6 Urine pH and urine flow

Passive reabsorption depends on whether drug is unionized and membrane-permeable in the tubular lumen. Weak bases become more ionized at low pH and are therefore less reabsorbed; their renal clearance can increase when urine becomes acidic. Weak acids become more ionized at high pH and are therefore less reabsorbed; their renal clearance can increase when urine is alkalinized. This is the logic behind urine alkalinization for some weak-acid overdoses.

Urine flow matters most when reabsorption is substantial. If a compound is hardly reabsorbed, forcing more urine cannot remove what was not returning anyway. If a compound is extensively reabsorbed, faster flow can reduce residence time for passive back-diffusion, increasing excretion [5].

2.7 Hepatic clearance: blood flow, binding, intrinsic clearance

The liver receives blood from the portal vein and hepatic artery. A useful adult average is $Q_H \approx 1.35$ L/min [5]. The classical well-stirred model summarizes hepatic blood clearance as

$$CL_H = \frac{Q_H \, f_{u,b} \, CL_{\text{int}}}{Q_H + f_{u,b} \, CL_{\text{int}}}.$$

The corresponding extraction ratio is

$$E_H = \frac{CL_H}{Q_H} = \frac{f_{u,b}\, CL_{\text{int}}}{Q_H + f_{u,b}\, CL_{\text{int}}}, \qquad F_H = 1 - E_H = \frac{Q_H}{Q_H + f_{u,b}\, CL_{\text{int}}}.$$

This single expression contains the high-extraction/low-extraction distinction:

$$f_{u,b} CL_{\text{int}} \gg Q_H \quad \Rightarrow \quad CL_H \approx Q_H \quad \text{flow-limited;}$$

$$f_{u,b} CL_{\text{int}} \ll Q_H \quad \Rightarrow \quad CL_H \approx f_{u,b} CL_{\text{int}} \quad \text{capacity/binding-limited.}$$

Question	High hepatic extraction	Low hepatic extraction
What limits CL_H?	Hepatic blood flow. The liver removes most drug delivered to it.	Intrinsic clearance and unbound fraction. Delivery is not the bottleneck.
Effect of increased blood flow	Often important.	Usually small.
Effect of enzyme induction	Systemic CL_H may change modestly, but oral F_H can change strongly.	CL_H often increases substantially.
Effect of enzyme inhibition	CL_H may be less sensitive than F_H; first pass may change.	CL_H and AUC can change strongly.
Effect of protein binding on total CL	Often limited for IV systemic clearance.	Often important for total clearance.
Clinical warning	Oral bioavailability may be low and variable due to first pass.	Total concentration changes can reflect binding and intrinsic clearance changes.

Table 1: The high/low extraction distinction is not a label to memorize. It predicts which physiological perturbation matters.

2.8 Phase I, Phase II, and transporters

The course transcript frames biotransformation as chemical conversion of parent drug into a different molecule. A simple teaching split is:

- **Phase I**: oxidation, reduction, hydrolysis. *CYP* enzymes are central, especially *CYP*3A, but not exclusive.
- **Phase II**: conjugation reactions such as glucuronidation, sulfation, acetylation, methylation, and glutathione conjugation.

This split is useful but incomplete unless transport is included. Hepatic clearance can require basolateral uptake into hepatocytes, metabolism or biliary excretion inside the cell, and canalicular efflux into bile. For polar compounds, uptake permeability or transporter capacity can be the rate-limiting step even when enzyme capacity is high. The modern question is therefore not only "which enzyme metabolizes it?" but:

> *Which concentration drives elimination, and what barrier separates measured plasma from the eliminating site?*

2.9 Induction and inhibition

For low-extraction hepatic drugs, induction of enzymes or transporters often increases CL_H, lowers AUC, and can shorten half-life if V_d is unchanged. Inhibition does the reverse. For high-extraction drugs, systemic hepatic clearance after IV dosing may be buffered by blood flow; however, oral exposure can still change because $F_H = 1 - E_H$ is sensitive to first-pass extraction.

This resolves an apparent paradox. A high-extraction drug can have flow-limited systemic clearance and yet strong oral bioavailability changes after enzyme inhibition, because the oral pathway asks how much survives the first pass. A small absolute change in E_H near one can be a large relative change in F_H.

2.10 First-pass metabolism

Oral bioavailability decomposes as

$$F = F_a F_G F_H,$$

where F_a is fraction absorbed into enterocytes, F_G is fraction escaping gut-wall metabolism or efflux back to lumen, and F_H is fraction escaping hepatic first pass. Module 2C returns to this expression in detail, but it belongs here too because F_H is an extraction statement:

$$F_H = 1 - E_H.$$

First pass is not a mysterious route separate from clearance. It is organ extraction applied before drug reaches systemic circulation.

2.11 Regulatory update: organ impairment

FDA's current renal impairment guidance was finalized in March 2024 and is intended to help sponsors decide when and how to study the effect of renal impairment on investigational-drug *PK* and dosing [8]. FDA's hepatic impairment guidance remains the 2003 final guidance and covers *PK* and, when relevant, *PD* study design and labeling in hepatic impairment [6]. These guidances convert the textbook mechanisms into development questions:

- Is renal or hepatic elimination a meaningful fraction of total clearance?
- Are active metabolites present?
- Does impaired organ function alter protein binding, transport, or nonrenal/nonhepatic pathways?
- Can exposure changes be translated into dose adjustment or contraindication language?
- Is the relevant endpoint total concentration, unbound concentration, exposure, response, or safety margin?

The important shift for students is that organ impairment studies are not ceremonial. They are how clearance mechanisms become label instructions.

3 2B: Protein Binding and Drug Distribution

3.1 Distribution has rate and extent

Drug distribution means reversible transfer between systemic circulation and tissues. Two questions must be separated:

- **Rate**: how fast does drug move from blood to tissue and back?
- **Extent**: how much drug resides outside the measured plasma or blood space at distribution equilibrium?

Blood perfusion, capillary permeability, tissue permeability, active transport, pH gradients, and membrane properties govern rate. Lipophilicity, ionization, plasma protein binding, tissue binding, red-cell partitioning, and body composition govern extent.

The course examples make this intuitive. Highly lipophilic drugs can distribute into adipose tissue, creating large apparent volumes and long persistence. Hydrophilic drugs or large molecules may remain mostly in plasma and extracellular water. The blood-brain barrier is a special permeability boundary: it is not enough for a drug to be in blood; it must cross a protected interface.

3.2 Volume of distribution is a proportionality

$$V_d = \frac{A_{\text{body}}}{C_p}.$$

If a drug is mostly in plasma, C_p is high relative to amount, so V_d is small. If a drug leaves plasma and binds extensively in tissue, C_p is low relative to amount, so V_d is large. V_d is therefore a statement about measurement and distribution, not anatomical size.

At steady state, a conceptual *PBPK* expression is

$$V_{ss} \approx V_p + \sum_T V_T K_{p,T},$$

where $K_{p,T}$ is a tissue-to-plasma partition coefficient. This expression is not meant to be memorized as a universal formula. It says that V_{ss} is built from tissue volumes weighted by tissue partitioning.

3.3 Protein binding: only unbound drug crosses and acts, but total drug is what we often measure

Plasma protein binding is usually rapid, reversible, and approximately linear over therapeutic concentrations, although exceptions are clinically important. The unbound fraction is

$$f_u = \frac{C_u}{C_p}.$$

The free-drug hypothesis says that unbound drug is the diffusible, pharmacologically active, filterable, and often metabolizable species. This is why f_u appears in filtration, hepatic clearance, in vitro–in vivo extrapolation, and interpretation of antimicrobial activity.

But the phrase "only free drug matters" can be misleading if it makes us ignore the system. Bound drug is a reservoir. Total concentration can change while unbound concentration remains almost unchanged, especially when clearance adapts to unbound concentration. Conversely, in nonlinear binding, hypoalbuminemia, uremia, critical illness, pregnancy, displacement interactions, or narrow-therapeutic-index drugs, unbound concentration may be the right measurement.

3.4 When protein binding matters

The distribution lecture emphasizes that plasma protein binding should be interpreted differently depending on extraction class [10]. In the low-extraction well-stirred approximation,

$$CL_H \approx f_{u,b} CL_{\text{int}}.$$

Increasing $f_{u,b}$ can increase total clearance. In high extraction,

$$CL_H \approx Q_H,$$

so systemic clearance is less sensitive to $f_{u,b}$, at least under the assumptions of perfusion-limited distribution and rapid binding equilibrium.

For steady-state IV infusion, the distinction between total and unbound concentration is essential. A change in protein binding can lower total concentration while leaving unbound concentration similar, because clearance and distribution are both referenced to unbound drug. This is one reason simple displacement-from-protein-binding stories often overpredict *DDI* risk.

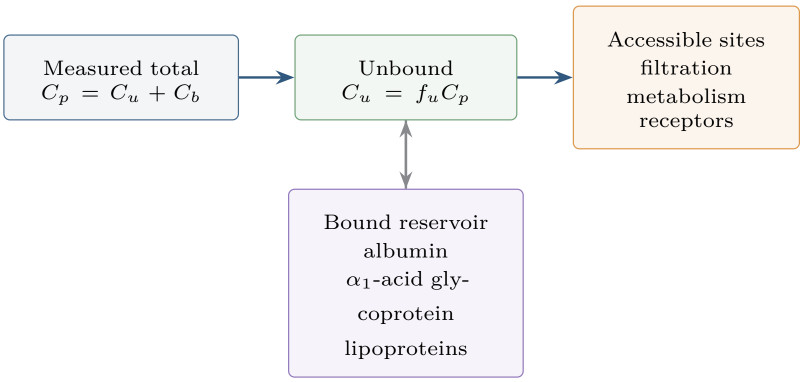

Figure 4: Protein binding links total and unbound concentration. The unbound concentration is often the pharmacological driver, but total concentration is what many assays report.

3.5 Species, methods, and nonlinear binding

The course spends time on measurement variability because this is where pharmacology becomes experimental rather than merely algebraic. Binding can differ across species; a human f_u cannot be assumed from rat or dog. Binding can differ across assay methods such as equilibrium dialysis, ultrafiltration, and microdialysis. It can differ with albumin source, serum composition, pH, temperature, and concentration. Nonlinear binding can appear when binding sites saturate, making f_u concentration-dependent.

This matters for translation:

– in vitro potency measured in protein-free medium may not predict in vivo potency;

– antimicrobial *MIC* can shift when serum is added;

– animal-to-human potency comparisons should often use unbound exposure;

– therapeutic drug monitoring may need unbound concentration for phenytoin, valproate, ceftriaxone, and selected critical-care cases.

3.6 Distribution is not homogeneous tissue partitioning

A visually important slide in the distribution deck warns that tissue partition coefficients are not always appropriate because they imply homogeneous tissue distribution. This is a subtle but modern point. If a drug is trapped in lysosomes, actively transported into hepatocytes, excluded from brain, or bound in a subcellular compartment, then a single K_p can be a useful empirical summary but a poor mechanism.

Azithromycin is a good example from the course: lysosomal trapping can create complex tissue distribution that cannot be reduced to "plasma times one partition coefficient" without losing the mechanism. In *PBPK* language, the question becomes whether tissue is perfusion-limited, permeability-limited, transporter-limited, pH-partition-limited, or binding-limited.

3.7 Jeong and Jusko: fractional distribution parameter fd

Jeong and Jusko's recent work is valuable because it challenges a hidden assumption behind many classical clearance and distribution formulas: rapid perfusion-limited exchange between vascular blood and tissue [3, 4]. They introduce a fractional distribution parameter f_d in *PBPK*-style models. The product

$$Q_T\, f_d$$

acts like an apparent distributional clearance into tissue. If $f_d = 1$, the tissue behaves more like a perfusion-limited compartment. If $f_d < 1$, only a fraction of delivered drug effectively crosses into tissue during a pass, reflecting permeability or uptake limitation.

Their outflow idea can be taught in words before equations:

> *Blood leaving an organ is a mixture of drug that did not distribute into tissue during that pass and drug returning from tissue.*

A compact version is

$$C_{\text{out}} = C_{\text{in}}(1 - f_d) + \frac{C_T}{K_p}\, R_b\, f_d.$$

This equation is not needed for routine noncompartmental analysis, but it is conceptually important. It says that flow Q is not always the right effective contact rate between blood and eliminating cells. Sometimes the relevant term is closer to Qf_d.

3.8 What this changes for a student

The classical low/high extraction teaching remains useful. The update is that the student should ask more mechanistic questions before applying it blindly:

- Is the eliminating enzyme or transporter inside a tissue compartment separated from blood by a permeability barrier?
- Does unbound plasma concentration equal unbound tissue concentration quickly enough?
- Is uptake into hepatocytes or kidney cells rate-limiting?
- Is measured K_p a steady-state partition coefficient, a terminal-phase tissue/plasma ratio, or a model-derived parameter?
- Does the organ have transit-time delay between inlet and outlet concentration?

This is the bridge from Module 2 to modern *PBPK*. The equations in introductory *PK* are not wrong; they are limiting cases of a richer transport-distribution-elimination system.

4 2C: Drug Absorption

4.1 Oral absorption is rate and extent

Oral drug delivery is common because it is convenient, scalable, and patient-friendly. But the oral route inserts a complicated system between dose and systemic exposure. The observed AUC, $C_{\max}$, and $T_{\max}$ after oral administration reflect:

$$\begin{gathered}\text{formulation} \rightarrow \text{dissolution} \rightarrow \text{GI transit} \rightarrow \text{permeability/transport} \\ \rightarrow \text{gut metabolism} \rightarrow \text{hepatic first pass} \rightarrow \text{systemic clearance.}\end{gathered}$$

Extent is summarized by F, but rate is summarized by k_a, $T_{\max}$, early exposure, or a mechanistic absorption model. A drug can have unchanged AUC but delayed $T_{\max}$; that may matter greatly for pain relief or sleep induction and less for chronic suppression of a biomarker.

4.2 The sink condition

The course overview says systemic absorption is favored because the body acts as a sink. This is a beautiful way to connect absorption with distribution and clearance. If systemic unbound concentration stays low because drug distributes into tissues and is eliminated, the concentration gradient across the gut wall is maintained:

$$J = P\,A\,(C_{\text{lumen},u} - C_{\text{blood},u})\,.$$

Here P is permeability and A is absorptive surface area. Absorption is not just a property of the gut wall. It depends on the downstream system keeping blood concentration lower than lumen concentration.

4.3 Dissolution-limited versus permeability-limited absorption

The Module 2C objectives ask the student to distinguish dissolution and permeability rate limitations. A practical classification is:

- **Dissolution-limited**: drug cannot enter solution fast enough. Particle size, salt form, formulation, bile salts, and food can matter.
- **Solubility-limited**: drug has poor aqueous solubility; concentration in GI fluid is capped.
- **Permeability-limited**: drug dissolves but crosses the intestinal barrier slowly.
- **Transit-limited**: drug has a limited window for absorption and moves past it.
- **First-pass-limited**: drug enters enterocytes or portal blood but is extracted by gut wall or liver before reaching systemic circulation.

ICH M9 formalizes the *BCS* language used in biowaivers. In vivo bioequivalence studies generally assess rate and extent using AUC and $C_{\max}$. For appropriate immediate-release products, high solubility and permeability classifications can sometimes support waiving in vivo studies [1]. For Module 2, the pedagogical value is simple: solubility and permeability are not interchangeable.

4.4 pH partition is useful, but not enough

The absorption transcript uses aspirin to warn against overreading pH-partition theory. Weak acids are more unionized in acidic gastric pH, so a naive pH-partition argument predicts substantial stomach absorption. But real oral absorption also depends on dissolution, surface area, residence time, unstirred water layer, gastric emptying, intestinal permeability, and blood flow. The small intestine often dominates because it has enormous surface area and longer effective absorptive opportunity.

The rule is:

> *pH and pKa tell us ionization; they do not by themselves tell us absorption.*

4.5 Gastric emptying and intestinal transit

Gastric emptying controls when dissolved or dissolving drug reaches the small intestine. Food can delay gastric emptying, change gastric pH, stimulate bile, increase splanchnic blood flow, alter viscosity, and change the migrating motor complex. For highly soluble, highly permeable drugs such as acetaminophen, food may delay $T_{\max}$ and onset without changing AUC much. For poorly soluble lipophilic drugs, food can increase solubilization and exposure.

The course's posaconazole example illustrates positive food effect: a lipophilic, poorly soluble drug can have substantially higher oral bioavailability under fed conditions. The underlying mechanism is not simply "food slows the stomach". It is a formulation-solubility-bile-lipid system.

4.6 Transporters and intestinal metabolism

Passive diffusion is only one absorption route. Uptake transporters such as *OATP*, *PEPT1*, bile-acid transporters, and monocarboxylate transporters can move drugs into enterocytes. Efflux transporters such as *P-gp*, *BCRP*, and *MRP* can pump drug back into the lumen. Transporter-mediated kinetics can be saturable, segment-dependent, species-dependent, genotype-dependent, and susceptible to *DDI*.

The absorption deck discusses digoxin and rifampin as a classic efflux-transporter example: induction of *P-gp* can reduce oral bioavailability for *P-gp* substrates. ICH M12, finalized at Step 4 in 2024, is the globally harmonized drug interaction guideline covering metabolic enzymes and transporters, including in vitro evaluation, clinical studies, predictive modeling, interpretation, and risk management [2]. This modern regulatory framing fits Module 2 exactly: absorption and clearance are not separable when gut transporters and enzymes sit in the same enterocyte.

4.7 Bioavailability decomposition

For an oral dose,

$$F = F_a F_G F_H.$$

The experimental absolute bioavailability is

$$F = \frac{AUC_{\text{oral}}/D_{\text{oral}}}{AUC_{\text{IV}}/D_{\text{IV}}}.$$

FDA's 2022 bioavailability guidance covers BA information for INDs, NDAs, and supplements and specifically addresses oral dosage forms, modified-release products, and systemic exposure measures where appropriate [7]. The course formula therefore maps directly into regulatory study design: an oral AUC is not merely an academic curve; it is evidence for formulation, food effect, dose proportionality, bridging, and label instructions.

4.8 Saturable first-pass metabolism

First-pass metabolism can be saturable. If gut or hepatic metabolism has a finite capacity, increasing dose may increase F because a smaller fraction is extracted. This produces nonlinear exposure:

$$AUC_{\text{oral}} = \frac{F(D)\,D}{CL}.$$

Here nonlinearity can enter through $F(D)$ even if systemic clearance after IV dosing looks linear over the same concentration range. This is one reason oral dose proportionality studies are so informative: they test the whole input-plus-clearance system.

5 Putting 2A, 2B, and 2C Together

5.1 AUC, Cmax, Tmax are fingerprints of different mechanisms

A concentration–time curve is not one parameter. Different parts of the curve report on different mechanisms:

- AUC: extent of systemic exposure, strongly controlled by FD/CL under linear conditions.
- $C_{\max}$: peak exposure, shaped by dose, F, absorption rate, distribution, and clearance.
- $T_{\max}$: timing of peak, strongly affected by absorption rate, gastric emptying, formulation, and elimination.
- terminal slope: often distribution-elimination mixture; not always pure clearance.
- early partial AUC: important for rapid-onset indications and food/formulation assessment.

The same AUC can hide very different early exposure; the same $C_{\max}$ can occur with different total exposure.

5.2 One oral dose, many possible explanations

Suppose a patient has unexpectedly low oral AUC. Module 2 gives a differential diagnosis:

- the drug did not dissolve;
- the drug degraded in the gut;
- gastric emptying or intestinal transit moved the drug past its absorption window;
- permeability was low;
- efflux transporters returned drug to lumen;
- gut-wall metabolism was high;
- hepatic first-pass extraction was high;
- systemic clearance was high;
- plasma protein binding or blood/plasma partition changed the measured reference concentration;
- sampling missed the peak or terminal phase.

The solution is not to memorize more parameters, but to ask which mechanism would change which observable.

5.3 A compact diagnostic table

Observation	Plausible mechanism	What to check next
Low oral AUC, normal IV CL	Poor F_a, low F_G, or low F_H	Absolute BA, mass balance, metabolite profile, transporter/enzyme data.
Food delays $T_{\max}$, unchanged AUC	Gastric emptying delay	Early exposure and onset-of-action relevance.
Food increases AUC	Solubilization, bile/lipid effects, formulation dependence	Fed/fasted BA, dissolution, lipophilicity, bile-salt sensitivity.
$CL_R \approx f_u GFR$	Filtration-dominated renal clearance	Renal impairment study relevance depends on f_e.
$CL_R > f_u GFR$	Net tubular secretion	Transporter inhibition/competition, renal *DDI*.
$CL_R < f_u GFR$	Net reabsorption	Urine pH, urine flow, lipophilicity, pKa.
High $CL_H \approx Q_H$	High extraction, flow-limited systemic clearance	First-pass, hepatic blood flow, route dependence.
Large V_d	Tissue binding/partitioning or low plasma concentration	Lipophilicity, tissue K_p, lysosomal trapping, RBC partition.
Total concentration changes but response does not	Protein binding shift with stable unbound exposure	Measure or model unbound concentration.

Table 2: Mechanism-oriented reading of common *PK* observations.

5.4 The Jeong/Jusko challenge to the classical picture

Classical clearance models are teaching masterpieces because they compress physiology into tractable equations. But Jeong and Jusko point out that the compression hides assumptions [4]:

- no relevant membrane barrier between vascular blood and eliminating tissue;
- instantaneous unbound plasma–tissue equilibrium;
- intrinsic clearance acts on a concentration that is treated as directly accessible from blood;
- transit-time delay between inlet and outlet is often not represented in closed form.

Their extended Rattle, Sieve, Tube, and Jar models can be read as modern counterparts of dispersion, series-compartment, parallel-tube, and well-stirred ideas. For this course, the philosophical lesson is enough:

> *Extraction is not only about enzyme capacity and blood flow; it is also about how drug physically reaches the eliminating site.*

This is especially relevant for transporter substrates, polar compounds, permeability-limited tissues, and *PBPK* applications.

6 Worked Examples

6.1 Is this renal drug filtered, secreted, or reabsorbed?

Suppose $f_u = 0.25$, $GFR = 120$ mL/min, and measured $CL_R = 75$ mL/min. Filtration alone predicts

$$f_u GFR = 0.25 \times 120 = 30 \text{ mL/min}.$$

Because $75 > 30$, net secretion must contribute. If a transporter inhibitor is coadministered, CL_R may fall, exposure may increase, and the *DDI* could be clinically relevant if renal clearance is a meaningful fraction of total clearance.

6.2 High extraction and first pass

Suppose $Q_H = 90$ L/h, $f_{u,b} CL_{\text{int}} = 810$ L/h. Then

$$E_H = \frac{810}{90 + 810} = 0.90, \qquad F_H = 0.10.$$

If an inhibitor halves $f_{u,b} CL_{\text{int}}$ to 405 L/h,

$$E_H = \frac{405}{90 + 405} = 0.818, \qquad F_H = 0.182.$$

Systemic hepatic clearance changes from 81 to 73.6 L/h, a modest decrease, but hepatic escape after oral dosing almost doubles. This is the high-extraction oral paradox: first-pass survival can be very sensitive even when IV systemic clearance is flow-limited.

6.3 Low extraction and protein binding

Suppose $Q_H = 90$ L/h, $f_{u,b} = 0.01$, and $CL_{\text{int}} = 900$ L/h. Then $f_{u,b} CL_{\text{int}} = 9$ L/h, so

$$CL_H = \frac{90 \times 9}{90 + 9} = 8.18 \text{ L/h}.$$

If disease doubles unbound fraction to $f_{u,b} = 0.02$, then $f_{u,b} CL_{\text{int}} = 18$, and

$$CL_H = \frac{90 \times 18}{90 + 18} = 15.0 \text{ L/h}.$$

Total clearance rises. Total concentration at the same dose may fall. The unbound concentration may not double and may even remain closer than expected after the system equilibrates. This is why total and unbound exposure need separate thinking.

6.4 Food effect interpretation

Suppose an oral product has unchanged AUC but delayed $T_{\max}$ and lower $C_{\max}$ after food. A reasonable first explanation is delayed gastric emptying and slower input, not reduced extent of absorption. For a chronic antihypertensive, this may not matter; for an analgesic intended for rapid relief, it may matter clinically. If food instead increases AUC fivefold for a lipophilic poorly soluble drug, the first suspicion shifts toward solubilization and bile/lipid-mediated dissolution or formulation effects.

7 How to Study This Module

7.1 What to memorize precisely

Memorize these:

$$\text{rate of elimination} = CL\,C, \qquad CL_B = QE, \qquad E = \frac{C_{\text{in}} - C_{\text{out}}}{C_{\text{in}}}.$$

$$\begin{aligned} CL_{\text{filtration}} &= f_u\,GFR, \\ CL_R > f_u\,GFR &\Rightarrow \text{secretion}, \\ CL_R < f_u\,GFR &\Rightarrow \text{reabsorption}. \end{aligned}$$

$$CL_H = \frac{Q_H f_{u,b}\,CL_{\text{int}}}{Q_H + f_{u,b}\,CL_{\text{int}}}, \qquad F = F_a F_G F_H.$$

$$V_d = \frac{A_{\text{body}}}{C_p}, \qquad f_u = \frac{C_u}{C_p}.$$

7.2 What not to memorize blindly

Do not say "high protein binding means low clearance" without asking extraction class and route. Do not say "weak acids absorb in stomach" without asking dissolution, surface area, residence time, and intestinal permeability. Do not say "clearance is metabolism" because renal excretion, biliary excretion, uptake transport, and blood flow can be equally important. Do not say "large V_d means drug goes everywhere"; it may mean plasma concentration is low because of a particular tissue or binding compartment.

7.3 A personal checklist

For every Module 2 problem, I will ask:

– Which concentration is referenced: plasma, blood, or unbound?

– Is the observation about rate, extent, or both?

– Is clearance renal, hepatic, biliary, metabolic, or total?

– Is the drug high extraction or low extraction in the relevant organ?

– Does protein binding change total concentration, unbound concentration, clearance, distribution, or several of these at once?

– Is oral exposure limited by F_a, F_G, F_H, or systemic CL?

– Are transporters or permeability barriers between measured concentration and the eliminating site?

– Would the label need renal impairment, hepatic impairment, food-effect, DDI, or dose-adjustment language?

8 Summary

8.1 One paragraph

Module 2 turns *ADME* from a mnemonic into a coupled physiological model. Clearance is the proportionality between concentration and elimination rate, but its value depends on the reference concentration and organ mechanism. Renal clearance is filtration plus secretion minus reabsorption, with $f_u GFR$ as the filtration benchmark. Hepatic clearance is shaped by blood flow, unbound fraction, intrinsic clearance, transport, and first pass, with high- and low-extraction drugs responding differently to induction, inhibition, binding, and flow. Distribution determines how amount becomes measured concentration, while protein binding separates total and unbound concentration. Absorption determines rate and extent of systemic input through dissolution, permeability, transporters, transit, food, gut metabolism, and hepatic first pass. Jeong and Jusko's f_d update reminds us that the classical equations are limiting cases: drug must physically reach tissues and eliminating cells before intrinsic clearance can matter. This is the conceptual spine for interpreting AUC, $C_{\max}$, $T_{\max}$, organ impairment, food effects, and *DDI* in later modules.

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